

PROJECT-BASED LEARNING – REVIEW 4

Enterprise Knowledge Intelligence Platform

Hybrid Relational, Document, and Vector Data Architecture

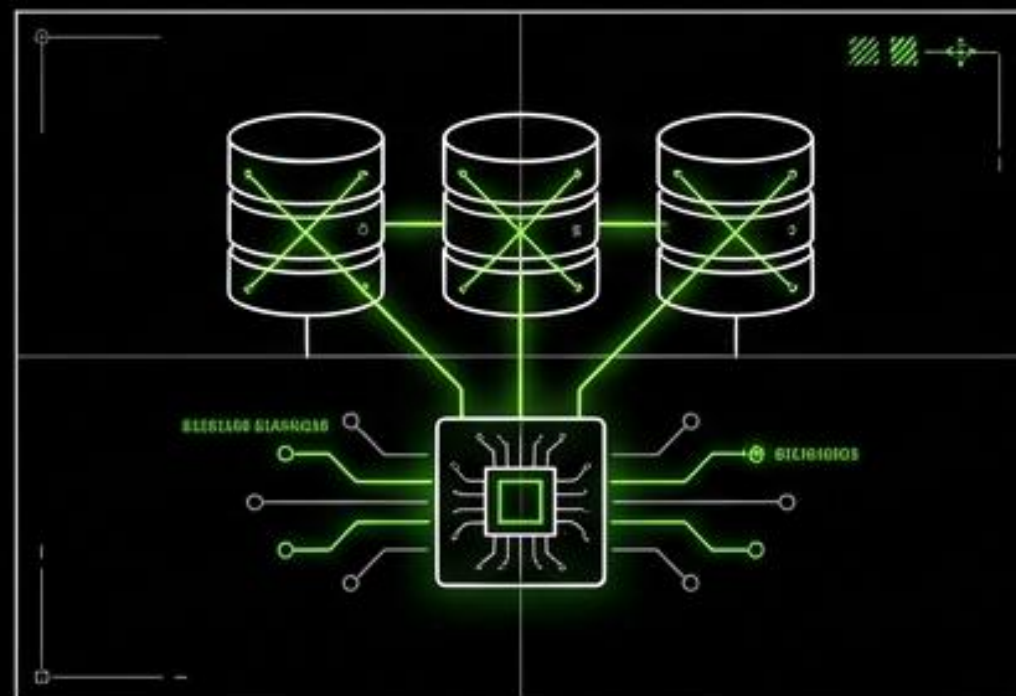
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Department of AI & Data Science

Academic Year: 2026-27

Project Objectives



Unify Organizational Knowledge

Consolidate disjointed files and organizational silos into a single, highly available polyglot repository (PostgreSQL, MongoDB, Qdrant).



Enable Hybrid Intelligence

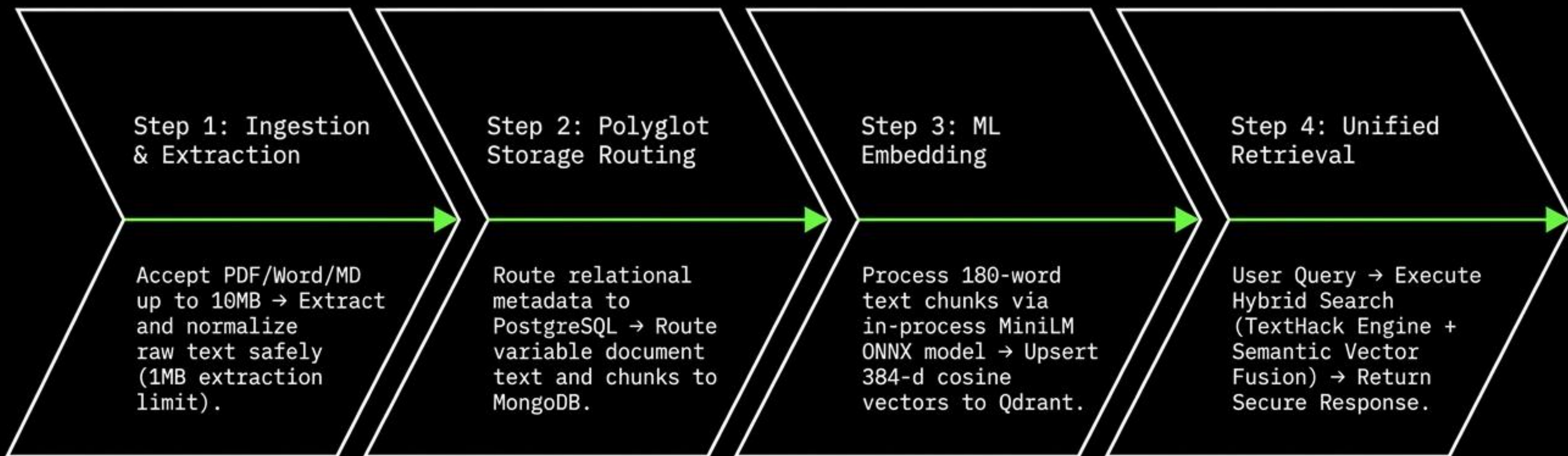
Retrieve complex documents by fusing exact keyword tracking, typo-tolerant (fuzzy) string matching, and deep semantic meaning.



Enforce Granular Security

Implement strict, document-level Role-Based Access Control (RBAC) so users only retrieve and view what they are explicitly permitted to see.

Project Development Approach



Project Implementation

Frontend

- React 19 SPA
- Vite 8 & React Router 7
- Permission-aware UI logic

Backend

- Spring Boot 3.4.2 (Java 21)
- In-process MiniLM ONNX ML Model
- DSA-3 TextHack Algorithm Engine

Database

- PostgreSQL 16 (Relational Metadata)
- MongoDB 7 (Document Content)
- Qdrant 1.12 (Vector Embeddings)

APIs

- RESTful Architecture
- JWT (HS256) Stateless Authentication
- Saga pattern compensating transactions

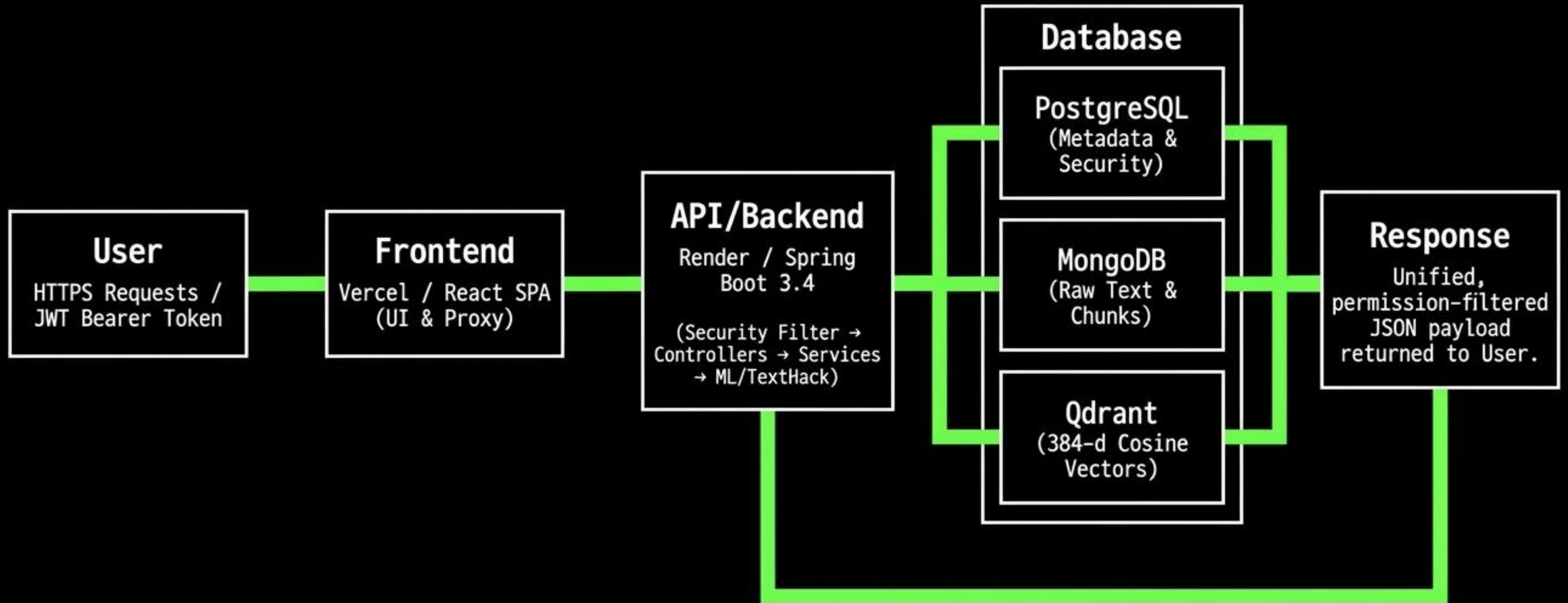
Development Environment

- Docker Compose (Full-stack containers)
- Maven & gcc 13
- Vitest & JUnit 5

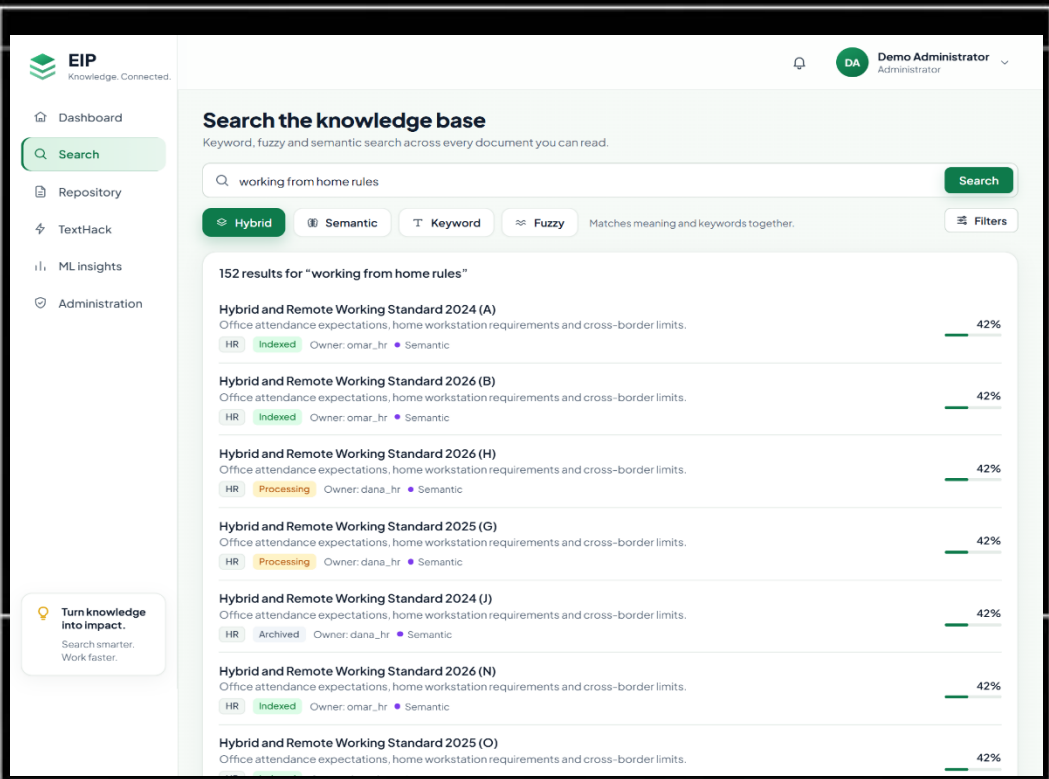
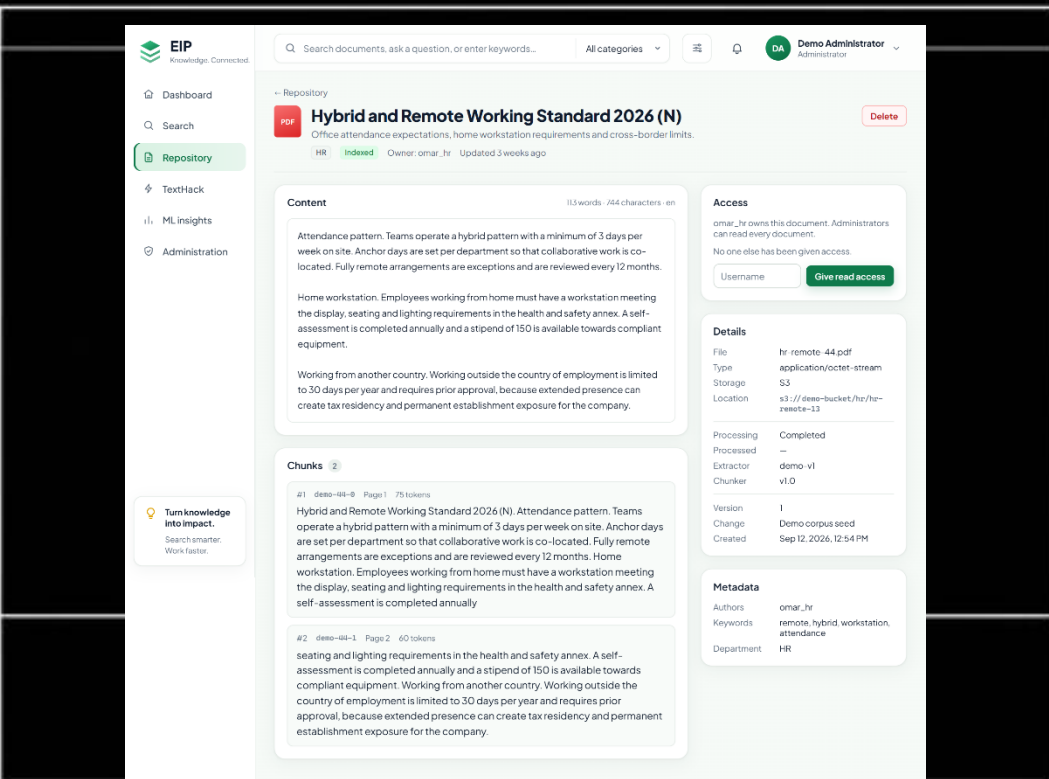
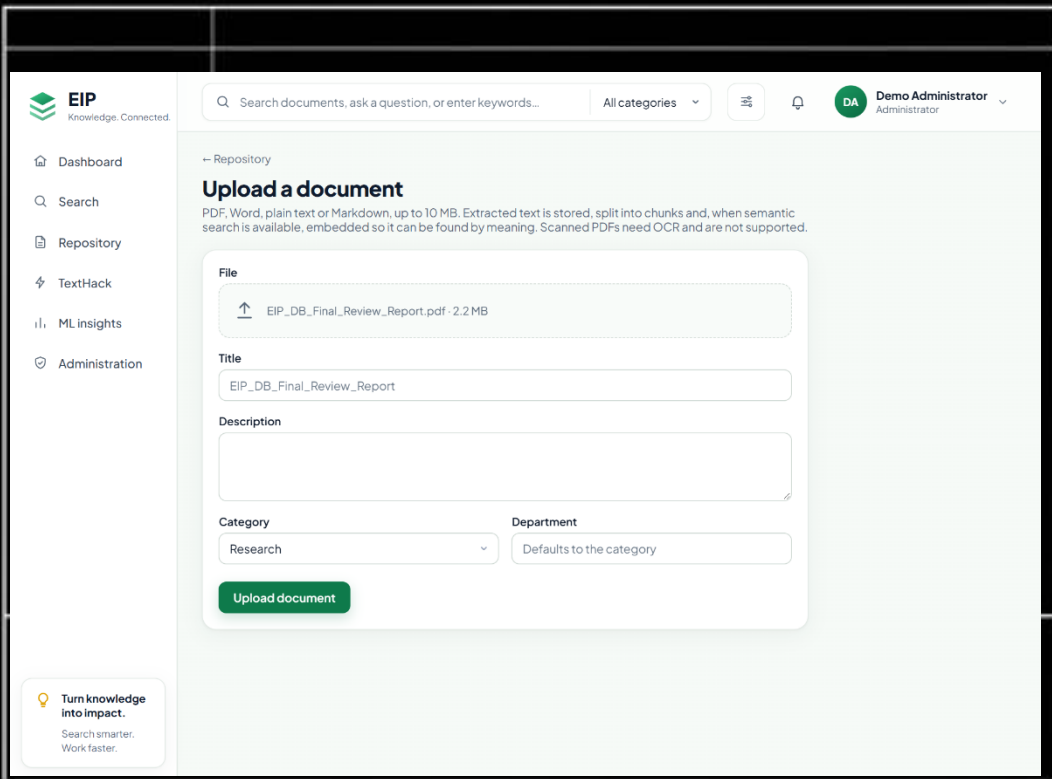
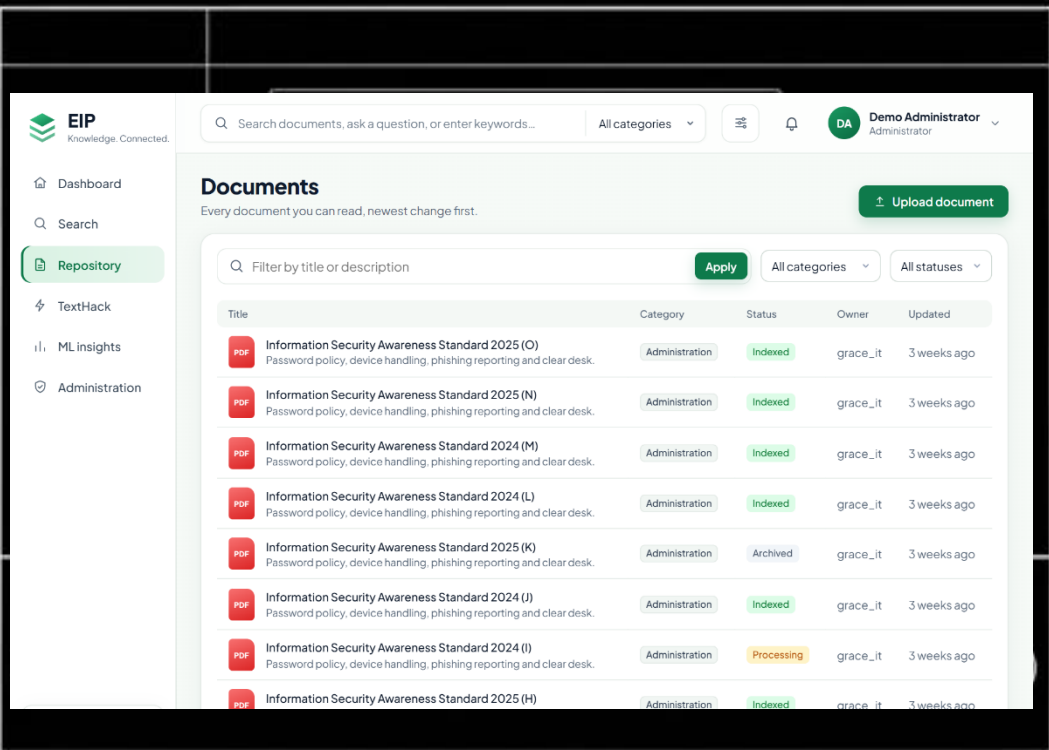
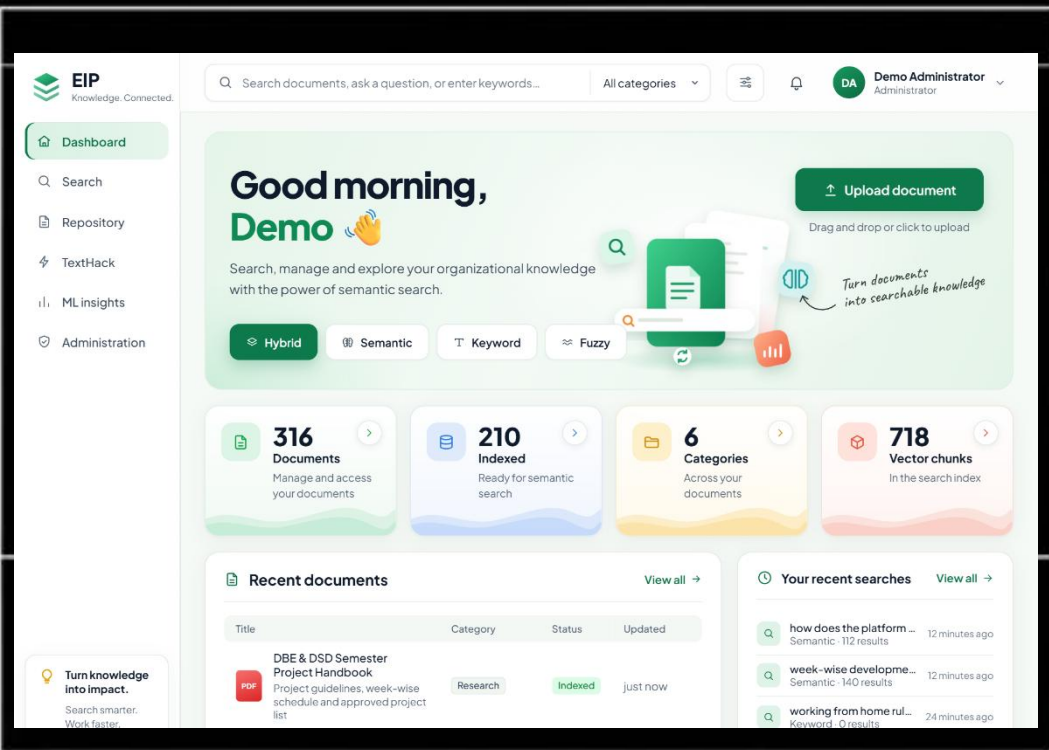
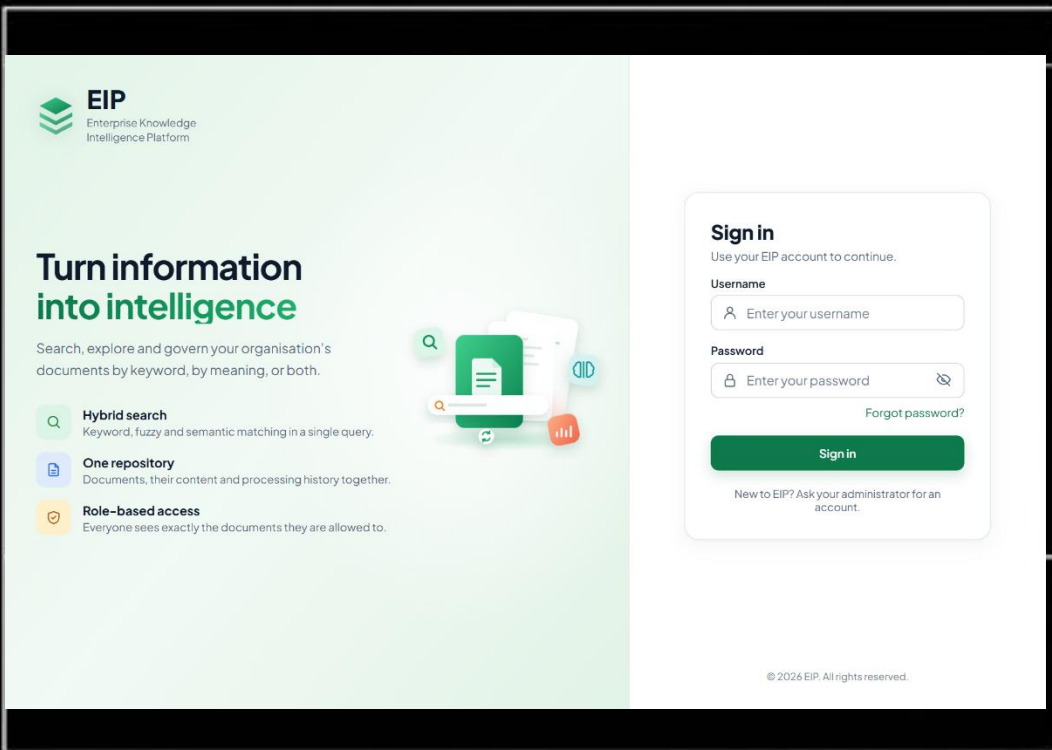
Deployment Constraints

- Vercel (UI Proxy)
- Render Free Tier (API)
- Tuned for severe 512 MB memory limit

System Architecture



Screens / Working Demonstration



Functional Results

	User Registration & Login Stateless JWT authentication backed by strict Role-Based Access Control (Admin, Manager, Employee).		Search & Filtering Unified hybrid search (0.4 keyword + 0.6 vector fusion) offering exact, fuzzy, and semantic modes.
	Database Operations Compensating-transaction (Saga) patterns keeping 3 distinct databases consistent without cross-system locking.		API Communication Secure REST API proxying with strict error masking and seamless JSON payload hydration.
	Add/Update/Delete Records Secure multi-part document upload with automated chunking (180 words/40 overlap) and cascaded deletion.		Report Generation Live ML Insights dashboard visualizing document classification and DBSCAN clustering metrics.

Testing Results

Test Case	Expected Result	Actual Result	Status
Login	Token generated & validated	Successful login & routing	[Pass]
Add Record	Extracted, chunked, synced to 3 DBs	Record added & vectorized	[Pass]
Update Record	READ permissions successfully modified	Record updated securely	[Pass]
Delete Record	Saga rollback across all 3 databases	Record deleted entirely	[Pass]
API Request	Fused search results returned returned < 1s	Correct response w/ matchedBy	[Pass]
Load Test	50 concurrent users, zero failures	28/28 checks cleared	[Pass]

Performance Analysis

0.4s – 0.9s

Response time

(Semantic Search on free tier constraint)

339

Number of test cases

(191 Backend + 148 Frontend assertions)

100%

Successful test cases

(339 / 339 passed)

0 Errors

Error rate

(Load tested at 169 requests/second)

< 50ms

Database response

(Indexed queries across polyglot trio)

446 MiB

API memory footprint

(Tuned JVM to fit inside 512MB free-tier limit)

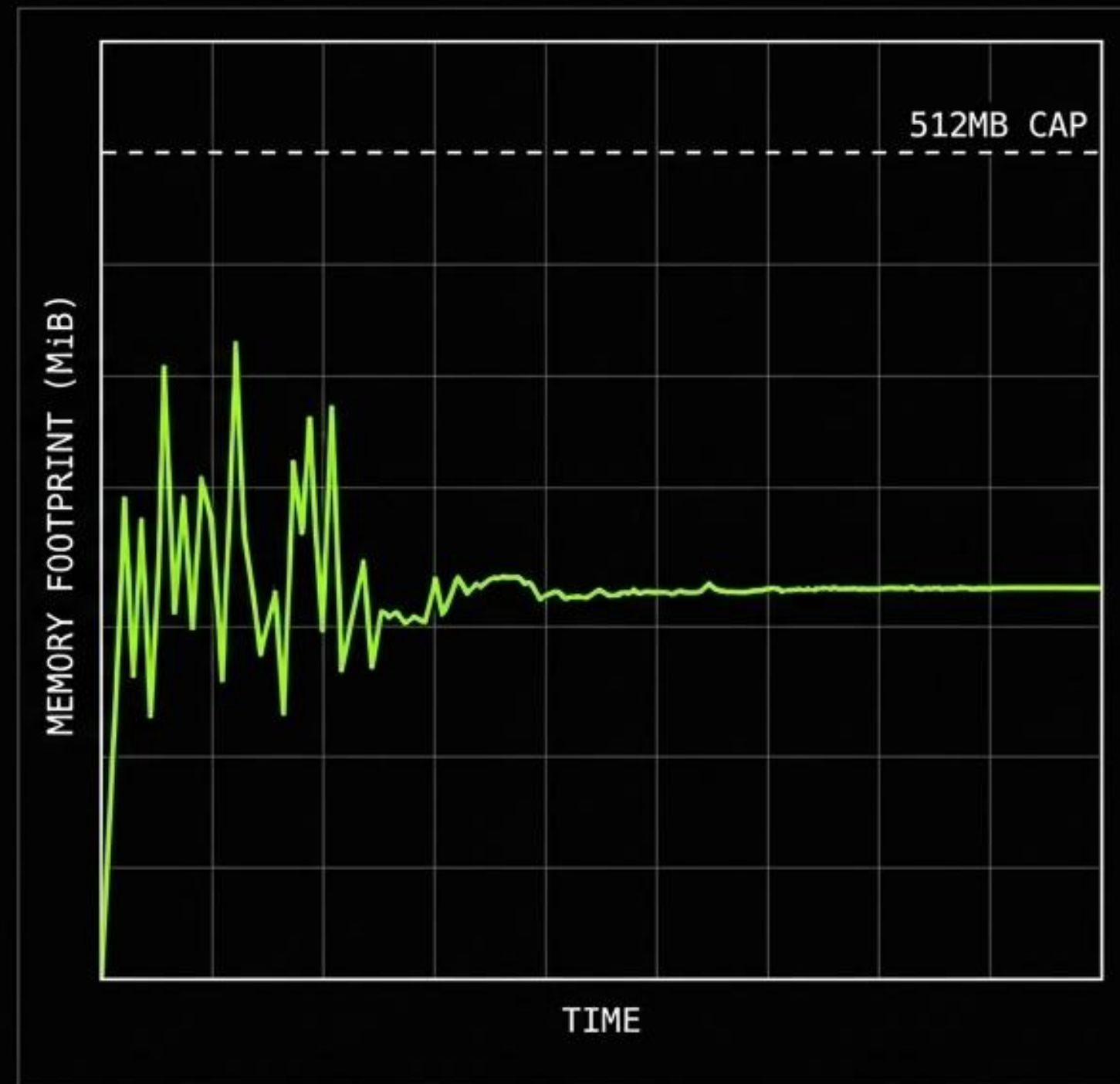
Results & Analysis

Robust System Stability: Achieved **100% test passing rates** across deep algorithmic evaluations and full-stack integration tests.

High Throughput, Low Resource: Successfully tuned the Java JVM (MALLOC_ARENA_MAX, 128MB heap) to process **169 requests/second** entirely within a highly restrictive **512MB** free-tier limit.

Algorithmic Efficiency: Custom TextHack algorithms demonstrated **15.5x faster processing** for multi-pattern searches (Aho-Corasick) compared to standard naive methods.

Retrieval Accuracy: MiniLM ONNX embedding significantly out-performed standard TF-IDF baselines in semantic retrieval (**nDCG@10 0.624** vs 0.371).



Conclusion & Future Scope

Conclusion

- Project objectives fully achieved.
- Delivered a production-ready, highly secure Enterprise Knowledge Intelligence Platform leveraging a live polyglot database architecture, custom advanced algorithms, and in-process machine learning.

Future Scope

- Implement Optical Character Recognition (OCR) to extract text from scanned PDFs.
- Expand MongoDB text indexing to allow full-body keyword search.
- Refactor monolithic backend into distinct microservices (Search vs. Ingestion) behind an API Gateway.
- Implement learned ranking weights for hybrid search fusion based on real user click data.